

Tezpur University
Autumn Semester Mid Examination - 2025

Course Code: CO434

Course Title: Image Processing

Full marks = 50

Time = 2 hours

(The figures in the right-hand margin indicate full marks for the questions)

1. Fill up the blanks in the following statements: 1×5=5
 - (a) The _____ colour has the longest wavelength amongst the colours in the visible light band.
 - (b) A disturbance that cause fluctuations in image pixel values is called _____.
 - (c) A noise that appears as bright spots in the lighter image is called _____.
 - (d) The splitting technique used in image segmentation can be represented using a _____ tree.
 - (e) Convolution in spatial domain is equivalent to _____ in frequency domain.
2. State whether the following statements are true or false: 1×5=5
 - (a) Max filter is an order statistic filter.
 - (b) The Laplacian is a first order derivative-based operator.
 - (c) An edge is a region of an image where there is a sudden change in the image intensity.
 - (d) The RGB colour model is a subtractive colour model is.
 - (e) Highpass filters perform image sharpening.
3. (a) What is a colour model? 1
 - (b) Give example of one hardware oriented colour model and one perceptive colour model. 2
 - (c) Describe the RGB colour model. 2
 - (d) Present hexadecimal codes for two all system safe RGB colours. 2
 - (e) How do you convert the CMY colour model to RGB colour model? 1
 - (f) How do you perform colour image smoothing? 2
4. (a) Given a 1-dimensional row image $f(x)$ find the first order and second order derivatives at location x . 1+2=3
 - (b) Derive the Laplacian filter for a 2-dimension image $f(x, y)$ based on the Laplacian operator: 3

$$\nabla^2 f(x, y) = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2}$$
 - (c) Incorporate the diagonal direction in the filter derived in (b) above and present the modified Laplacian filter. 1
 - (d) Apply the Laplacian filter derived in (c) above to the image given below and produce the sharpened output image. Compute the modified pixel values for the two central pixels only ignoring the boundary pixels. 3

2	6	3	7
9	12	8	3
3	7	3	5